MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3VACTXLA01 – CHEMICAL PROCESSING IN TEXTILES

Duration: 1.5Hrs **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
Answer All Questions
Each question carries 1 mark

1. Singeing is the process of removing [U] [2]
   a) Sizing agents b) Natural waxes c) Loose fibers d) Color
2. Dope dyeing is the method used for [U] [3]
   a) Yarns b) Finished fabrics c) Polymer solutions d) Fibers
3. Basic dyes are primarily used for [U] [3]
   a) Cotton b) Wool c) Acrylic d) Silk
4. Which dye class forms a strong chemical bond with cotton fibers? [U] [3]
   a) Direct dyes b) Reactive dyes c) Acid dyes d) Basic dyes
5. The mechanical process of smootheing fabric using rollers is [U] [2]
   a) Embossing b) Moire c) Calendering d) Mercerization
6. The color-bearing part of a dye molecule is the [U] [3]
   a) Chromophore b) Auxochrome c) Binder d) Mordant
7. Discharge printing works by [U] [4]
   a) Applying color b) Removing color c) Resisting color d) Transferring color
8. The printing method using engraved copper rollers is [U] [4]
   a) Block printing b) Roller printing c) Screen printing d) Stencil printing
9. Mercerisation of cotton increases its [U] [2]
   a) Luster b) Shrinkage c) Roughness d) Weight
10. What type of dyes are used in transfer printing? [U] [4]
    a) Natural dyes b) Mordants c) Disperse dyes d) Pigment dyes

[1 x 10 = 10]

Part B
Fill in the Blanks Questions
Answer All Questions
Each question carries 1 mark

11. The dye class also known as cationic dyes is _____ dyes. [U] [3]
12. Transfer printing is also known as _____ printing. [U] [4]
13. The finish that is applied to fabrics to prevent shrinkage during laundering is known as a _____ finish. [U] [2]
14. Reactive dyes are primarily used on _____ fibers. [U] [3]
15. _____ finish is designed to last for a significant portion of a product's life. [U] [1]
16. The removal of sericin gum from silk is called _____. [R] [2]
17. In pigment printing, the chemical used to fix the pigment to the fabric is a _____. [U] [3]

18. The oldest and simplest printing method uses carved wooden _____. [U] [4]
19. A dye that is insoluble and needs an alkaline solution is a _____ dye. [U] [3]
20. _____ technology uses an energized gas to modify a fabric's surface. [U] [2]

[1 x 10 = 10]

Part C

Answer in One Word or One Sentence
Answer All Questions
Each question carries 1 mark

21. State another name for Basic/Routine/General finishes. [U] [1]
22. Give the chemical name for caustic soda used in mercerization. [U] [2]
23. Who invented the first synthetic dye, basic mauve, in 1856? [U] [3]
24. What is the name of the fabric defect that can occur if tentering is not done properly? [U] [2]
25. Identify the tool used in screen printing to spread the colour paste evenly. [U] [4]
26. Give an example of a home-equivalent for the calendaring process. [U] [2]
27. Which printing style uses a chemical to remove a design from a pre-dyed fabric? [U] [4]
28. What machine dyes fabric in open width between two rollers? [U] [3]
29. Identify the dyeing method used to achieve a single, even color throughout a blended fabric. [U] [3]
30. Which printing method works similarly to an inkjet printer? [U] [4]

[1 x 10 = 10]

Part D
Short Answer Questions
Answer any five Questions
Each question carries 2 marks

31. Compare the main advantage of digital printing with conventional printing methods. [U] [4]
32. Explain the purpose of an antistatic finish. [U] [2]
33. What is the purpose of dope dyeing? [U] [3]
34. Differentiate between union dyeing and cross dyeing. [U] [3]
35. Explain the principle of discharge printing. [U] [4]
36. Classify textile finishes according to their life of a finish. [U] [1]

[2 x 5 = 10]

Part E
Long Answer Questions
Answer any two Questions
Each question carries 5 marks

37. Explain the main benefits of using plasma technology in textile finishing. [U] [2]
38. What is a key advantage of digital printing? [U] [4]
39. Explain the principle of resist printing and describe how it is used to create a design. [U] [4]
40. Discuss the various methods of yarn dyeing and explain the advantages of two of them. [U] [3]

[5 x 2 = 10]